Exp No:18 Experiment to understand KNN algorithm for a given data set

Description: Understand the KNN algorithm for the dataset given.

Code:

```
import numpy as np
import pandas as pd
df=pd.read_csv('/content/Iris.csv')
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 5 columns):
 #  Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0  sepal_length  150 non-null    float64
 1  sepal_width   150 non-null    float64
 2  petal_length  150 non-null    float64
 3  petal_width   150 non-null    float64
 4  species      150 non-null    object
dtypes: float64(4), object(1)
memory usage: 6.0+ KB
```

```
df.species.value_counts()
```

	count
species	
Iris-setosa	50
Iris-versicolor	50
Iris-virginica	50

```
dtype: int64
```

```
df.head()
```

	sepal_length	sepal_width	petal_length	petal_width	species
0	5.1	3.5	1.4	0.2	Iris-setosa

	sepal_length	sepal_width	petal_length	petal_width	species
1	4.9	3.0	1.4	0.2	Iris-setosa
2	4.7	3.2	1.3	0.2	Iris-setosa
3	4.6	3.1	1.5	0.2	Iris-setosa
4	5.0	3.6	1.4	0.2	Iris-setosa

```
features=df.iloc[:,:-1].values

label=df.iloc[:,4].values

from sklearn.model_selection import train_test_split

from sklearn.neighbors import KNeighborsClassifier

xtrain, xtest, ytrain, ytest = train_test_split(features, label, test_size=0.2, random_state=42)


model_KNN = KNeighborsClassifier(n_neighbors=5)

model_KNN.fit(xtrain, ytrain)

KNeighborsClassifier()

print(model_KNN.score(xtrain,ytrain))

print(model_KNN.score(xtest,ytest))

0.9666666666666667

1.0

from sklearn.metrics import confusion_matrix

confusion_matrix(label,model_KNN.predict(features))

array([[50,  0,  0],

    [ 0, 47,  3],

    [ 0,  1, 49]])
```

Result:

Understood the KNN algorithm for the given dataset.